

Energy-efficient Analog Integrated Circuits

Sensor Interface – Exercise 2

Assignment

Gabriele Atzeni

Prof. Taekwang Jang

1. Consider the amplifier depicted in Fig. 1, where $IRN = 5 \mu V_{RMS}$ is the input-referred noise of the open-loop amplifier. Consider $C_{IN} = 10 \text{ pF}$ and $C_F = 1 \text{ pF}$ and $R_F = 10 \text{ G}\Omega$.
 - a. Calculate the signal gain assuming the amplifier open-loop gain $A_{OL} \rightarrow \infty$.
 - b. What is the noise gain A_n ?
 - c. What is the input-referred noise of the closed-loop amplifier?
 - d. Calculate the NEF of the open-loop amplifier and closed-loop amplifier assuming the current consumption is $2 \mu A$ and the signal bandwidth is 10 kHz .

Discuss the effect of the parasitic input capacitance introduced by the input transistors of the amplifier on noise and NEF.

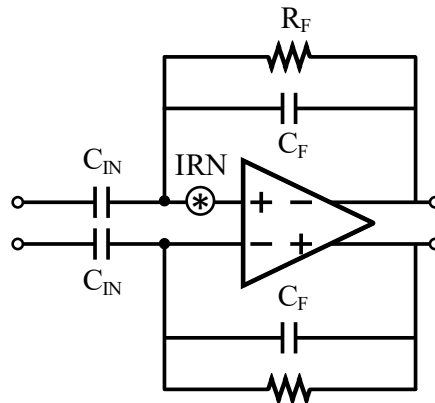


Fig. 1: Amplifier with capacitive feedback

2. Fig. 2 shows the schematic of an inverter-based amplifier with its common-mode feedback circuit. Assume $V_B = V_{DD}/2$ and $V_{DD} = 1.2 \text{ V}$. How would you choose the threshold voltage of M_1 and M_2 compared to M_3 and M_4 so that M_3 and M_4 operate as cascode devices?

3. Design the amplifier to meet the following specifications and explain your design strategy carefully:
 - Open-loop gain $A_{OL} > 40$ dB
 - Closed-loop gain $A_{CL} = 20$ dB
 - Bandwidth: 100 Hz – 15 kHz
 - Load capacitance $C_L = 5$ pF
 - Phase Margin $PM > 60^\circ$
 - Common-mode feedback phase margin $> 60^\circ$
 - Power consumption < 500 nW
 - Input-referred noise $IRN < 20 \mu V_{RMS}$
 - $NEF < 3$
4. Assume $R_F = 10$ G Ω and $C_{IN} = 5$ pF. Note: you need to use the diffstbprobe component from analogLib to run the stability analysis (stb) of a fully differential amplifier. Refer to Fig. 3 to check the probe configuration for the capacitive feedback and common-mode feedback. An ideal common-mode feedback circuit is available at /home/eeaic_04fs26/ExerciseTools. You

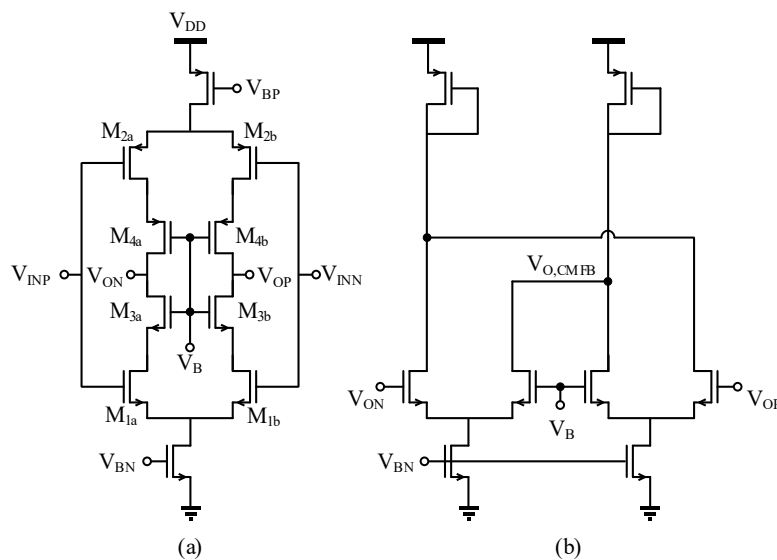


Fig 2: Inverter-based cascode amplifier (a), and its common-mode feedback circuit (b)

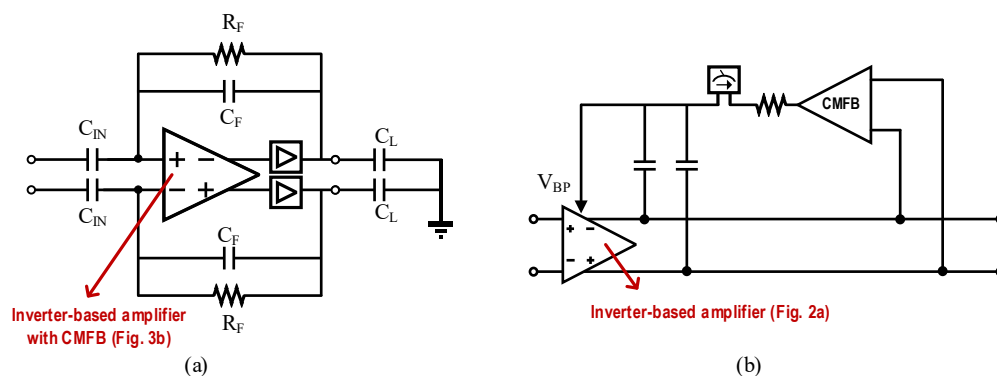


Fig. 3: Probe configuration for stability analysis of the capacitive feedback loop (a) and common-mode feedback loop (b).

can use `cp -r /home/eeaic_04fs26/ExerciseTools/ideal_cmfb <your cockpit folder>/dfii/<your library>` to copy the ideal model in your working library. Alternatively, you can design the common-mode feedback circuit using the topology in Fig. 2a. Discuss qualitatively the effect of a mismatch in the width of M_{1a} and M_{1b} . Verify your answer by introducing a 5% mismatch in simulation.

5. Assume your inverter-based amplifier is used as the first stage (CTA1) of the amplifier chain depicted in Fig. 4. Add a second stage (CTA2) and set C_{IN1} and C_{F1} to achieve a closed-loop gain of 5 for the first stage. Assume the load capacitance of the second stage is 100 fF. How should you set the current consumption of CTA2 to minimize the overall NEF?

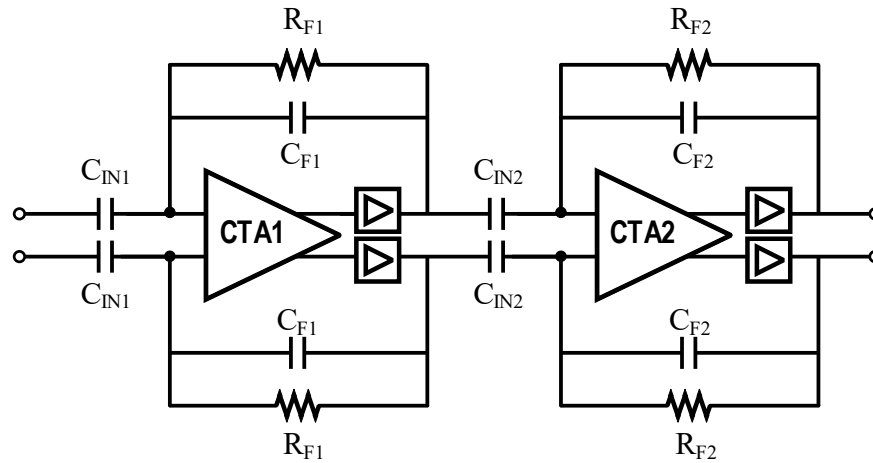


Fig. 4: Amplifier chain consisting of two inverter-based amplifiers with capacitive feedback